

AI Interview Questions & One-Line Answers

Low to Medium Expertise • 2026 Edition • LLMs, GenAI, RAG, Agents, MCP, MLOps, Security & AI Architecture

How to use: Read the question, answer aloud in your own words, then use the one-line answer as a checkpoint. The bank is designed for AI/GenAI interviews ranging from fundamentals to practical engineering discussions.

Total questions: 234

0. Current 2026 AI Interview Focus

Q. What AI topics are especially relevant in 2026 interviews?

Answer: Expect questions on LLMs, RAG, agents, tool calling, MCP, evaluation, AI security, governance, and production LLMOps.

Q. Why are AI agents a major interview topic?

Answer: Agents combine LLM reasoning with tools, state, orchestration, and actions, making reliability and safety important engineering concerns.

Q. Why is MCP increasingly relevant?

Answer: MCP standardizes how AI applications connect to external tools and context, making integrations more reusable.

Q. What is the key security principle for agent tools?

Answer: Treat tools as privileged capabilities and apply least privilege, validation, isolation, and approval to sensitive actions.

Q. What is the difference between RAG and agents?

Answer: RAG primarily supplies external knowledge, while agents can use tools and iterative workflows to accomplish goals.

Q. What is the difference between an LLM application and an agentic application?

Answer: An LLM application may make a direct model call, while an agentic application can dynamically decide steps and invoke tools.

1. AI & Machine Learning Fundamentals

Q. What is Artificial Intelligence (AI)?

Answer: AI is the field of building systems that perform tasks requiring capabilities such as reasoning, perception, learning, or decision-making.

Q. What is Machine Learning (ML)?

Answer: ML is a subset of AI where models learn patterns from data to make predictions or decisions.

Q. What is Deep Learning?

Answer: Deep learning uses multi-layer neural networks to learn complex representations from large amounts of data.

Q. What is Generative AI?

Answer: Generative AI creates new content such as text, images, audio, video, or code from learned patterns.

Q. What is the difference between AI, ML and Deep Learning?

Answer: AI is the broad field; ML is learning from data; deep learning is ML based mainly on multi-layer neural networks.

Q. What is supervised learning?

Answer: A model learns from labeled input-output examples.

Q. What is unsupervised learning?

Answer: A model discovers patterns or structure in unlabeled data.

Q. What is reinforcement learning?

Answer: An agent learns actions through rewards and penalties from an environment.

Q. What is classification?

Answer: Classification predicts a discrete category such as spam or not spam.

Q. What is regression?

Answer: Regression predicts a continuous numerical value such as price or demand.

Q. What is clustering?

Answer: Clustering groups similar data points without predefined labels.

Q. What is overfitting?

Answer: Overfitting occurs when a model memorizes training data and performs poorly on unseen data.

Q. What is underfitting?

Answer: Underfitting occurs when a model is too simple to capture important patterns in the data.

Q. What is a feature?

Answer: A feature is an input variable used by a machine-learning model.

Q. What is a label?

Answer: A label is the target value a supervised model is trained to predict.

Q. What is train-validation-test split?

Answer: Training fits the model, validation helps tune it, and testing estimates final generalization.

Q. What is data leakage?

Answer: Data leakage occurs when information unavailable at prediction time improperly enters training or evaluation.

Q. What is normalization?

Answer: Normalization rescales numerical values to a common range or distribution.

Q. What is standardization?

Answer: Standardization typically transforms a feature to have mean zero and standard deviation one.

Q. What is a model parameter?

Answer: A parameter is a value learned by the model during training.

Q. What is a hyperparameter?

Answer: A hyperparameter is configured before or around training, such as learning rate or batch size.

Q. What is inference?

Answer: Inference is the process of using a trained model to produce an output for new input.

2. Neural Networks & Transformers

Q. What is a neural network?

Answer: A neural network is a layered mathematical model that learns transformations from inputs to outputs.

Q. What is an activation function?

Answer: It introduces non-linearity so neural networks can learn complex relationships.

Q. What is backpropagation?

Answer: Backpropagation computes gradients of the loss with respect to model parameters.

Q. What is gradient descent?

Answer: Gradient descent updates parameters in the direction that reduces the training loss.

Q. What is a loss function?

Answer: A loss function measures how different a model's prediction is from the desired target.

Q. What is an epoch?

Answer: An epoch is one complete pass through the training dataset.

Q. What is batch size?

Answer: Batch size is the number of training examples processed before an update.

Q. What is a learning rate?

Answer: The learning rate controls the size of parameter updates during optimization.

Q. What is an embedding?

Answer: An embedding is a numerical vector representation that captures semantic or learned relationships.

Q. What is NLP?

Answer: Natural Language Processing enables computers to process and reason about human language.

Q. What is a token?

Answer: A token is a unit of text processed by a language model, such as a word piece, character, or symbol.

Q. What is tokenization?

Answer: Tokenization converts text into the tokens a model can process.

Q. What is a Transformer?

Answer: A Transformer is a neural architecture based heavily on attention mechanisms and parallel sequence processing.

Q. Why are Transformers important for GenAI?

Answer: They scale effectively, capture long-range relationships, and support efficient large-scale training.

Q. What is attention?

Answer: Attention lets a model assign different importance to parts of an input when producing an output.

Q. What is self-attention?

Answer: Self-attention lets tokens attend to other tokens within the same sequence.

Q. What are query, key and value in attention?

Answer: They are learned projections used to determine which information each token should attend to and retrieve.

Q. What is multi-head attention?

Answer: It runs multiple attention mechanisms in parallel so the model can capture different relationships.

Q. What is positional information?

Answer: It gives a Transformer information about token order because attention alone is not inherently sequential.

Q. What is encoder vs decoder?

Answer: Encoders primarily build representations from input, while decoders are designed to generate output autoregressively.

Q. What is an encoder-only model?

Answer: It mainly produces contextual representations and is well suited to understanding tasks.

Q. What is a decoder-only model?

Answer: It predicts the next token and is widely used for generative LLMs.

Q. What is an encoder-decoder model?

Answer: It uses an encoder to represent input and a decoder to generate output, useful for sequence-to-sequence tasks.

Q. What is an LLM?

Answer: A Large Language Model is a neural language model trained on very large text corpora to predict and generate tokens.

Q. What is a foundation model?

Answer: A foundation model is broadly pretrained and can be adapted to many downstream tasks.

Q. What is context window?

Answer: The context window is the amount of input and generated context a model can process in one interaction.

Q. What is temperature?

Answer: Temperature adjusts randomness in token selection, with lower values generally producing more deterministic outputs.

Q. What is top-p sampling?

Answer: Top-p sampling limits token selection to the smallest probability mass whose cumulative probability reaches p.

Q. What is hallucination?

Answer: A hallucination is an output that is unsupported, fabricated, or factually incorrect.

3. LLM Training & Adaptation

Q. What is pretraining?

Answer: Pretraining learns general language patterns from a very large corpus, often through next-token prediction.

Q. What is next-token prediction?

Answer: The model learns to predict the most likely next token given previous tokens.

Q. What is self-supervised learning?

Answer: The training signal is generated from the data itself rather than requiring manually assigned labels.

Q. What is instruction tuning?

Answer: Instruction tuning trains a model on examples of instructions paired with desired responses.

Q. What is supervised fine-tuning (SFT)?

Answer: SFT adapts a pretrained model using labeled input-output examples.

Q. What is RLHF?

Answer: Reinforcement Learning from Human Feedback uses human preferences to improve model behavior.

Q. What is DPO?

Answer: Direct Preference Optimization learns from preference pairs without requiring a separate reinforcement-learning loop.

Q. What is PEFT?

Answer: Parameter-Efficient Fine-Tuning adapts a small subset or low-rank representation of model parameters.

Q. What is LoRA?

Answer: LoRA adds trainable low-rank matrices while keeping most pretrained weights frozen.

Q. What is QLoRA?

Answer: QLoRA combines quantized base-model weights with LoRA adapters to reduce fine-tuning memory requirements.

Q. When should you fine-tune instead of using RAG?

Answer: Fine-tuning is useful for changing behavior, style, task performance, or domain patterns rather than simply supplying changing facts.

Q. When should you use RAG instead of fine-tuning?

Answer: Use RAG when the model needs grounded access to external or frequently changing knowledge.

Q. What is catastrophic forgetting?

Answer: It is the loss of previously learned capabilities after adapting a model to new data.

Q. What is quantization?

Answer: Quantization represents model values with lower precision to reduce memory use and often improve inference efficiency.

Q. What are INT8 and INT4?

Answer: They are low-bit numerical representations commonly used to reduce model memory and computation costs.

4. Prompt Engineering

Q. What is prompt engineering?

Answer: It is the practice of designing inputs and instructions that reliably guide a model toward the desired behavior.

Q. What is zero-shot prompting?

Answer: Zero-shot prompting asks a model to perform a task without providing examples.

Q. What is one-shot prompting?

Answer: One-shot prompting provides one example before asking the model to perform the task.

Q. What is few-shot prompting?

Answer: Few-shot prompting provides several examples to demonstrate the desired task or format.

Q. What is a system instruction?

Answer: It defines high-level behavior, constraints, role, or policies for the model.

Q. What is role prompting?

Answer: Role prompting frames the model as a particular expert or type of assistant.

Q. What is chain-of-thought prompting?

Answer: It is prompting intended to improve multi-step reasoning; production systems should generally avoid exposing private reasoning traces.

Q. What is structured output?

Answer: Structured output constrains responses to a defined schema such as JSON.

Q. What is function calling/tool calling?

Answer: It lets a model request a predefined function with structured arguments rather than directly performing the operation itself.

Q. What is prompt injection?

Answer: Prompt injection is an attempt to manipulate an AI system through malicious or conflicting instructions in input data.

Q. What is indirect prompt injection?

Answer: It occurs when malicious instructions are embedded in retrieved documents, webpages, files, or tool outputs.

Q. How do you reduce prompt injection risk?

Answer: Separate trusted instructions from untrusted data, constrain tools, validate outputs, and require approval for sensitive actions.

Q. What is prompt chaining?

Answer: Prompt chaining breaks a complex task into multiple sequential model calls.

Q. What is ReAct?

Answer: ReAct is an agent pattern that combines reasoning-oriented planning with actions and observations from tools.

Q. What is deterministic prompting?

Answer: It aims for repeatable outputs through stable prompts and low or controlled sampling randomness.

5. RAG & Vector Search

Q. What is RAG?

Answer: Retrieval-Augmented Generation retrieves relevant external information and supplies it to a model before generation.

Q. Why use RAG?

Answer: RAG can ground answers in enterprise or current knowledge without retraining the base model.

Q. What are the main RAG steps?

Answer: Ingestion, chunking, embedding/indexing, retrieval, optional reranking, context assembly, and generation.

Q. What is document ingestion?

Answer: It is the process of loading, parsing, cleaning, and preparing source documents for retrieval.

Q. What is chunking?

Answer: Chunking splits documents into smaller units that can be retrieved and supplied to a model.

Q. Why is chunking important?

Answer: Poor chunking can reduce retrieval relevance or waste context with irrelevant content.

Q. What is an embedding model?

Answer: It converts text or other content into vectors that represent semantic information.

Q. What is a vector database?

Answer: It stores and searches vector representations efficiently, often with metadata filtering.

Q. What is semantic search?

Answer: Semantic search retrieves content based on meaning rather than only exact keyword overlap.

Q. What is keyword search?

Answer: Keyword search retrieves documents using lexical term matching such as BM25.

Q. What is hybrid search?

Answer: Hybrid search combines lexical and semantic retrieval to improve coverage and relevance.

Q. What is BM25?

Answer: BM25 is a popular lexical ranking algorithm based on term frequency, document frequency, and document length.

Q. What is cosine similarity?

Answer: Cosine similarity measures the angle-based similarity between vectors.

Q. What is ANN search?

Answer: Approximate Nearest Neighbor search finds similar vectors efficiently without exhaustively comparing every vector.

Q. What is reranking?

Answer: Reranking applies a stronger relevance model to a smaller set of initially retrieved results.

Q. What is metadata filtering?

Answer: It restricts retrieval using attributes such as tenant, date, document type, or access level.

Q. What is top-k retrieval?

Answer: Top-k retrieval returns the k highest-ranked candidate results.

Q. What is query rewriting?

Answer: Query rewriting transforms the user's question into a retrieval-friendly form.

Q. What is multi-query retrieval?

Answer: It generates multiple related queries to improve recall for ambiguous or complex questions.

Q. What is HyDE?

Answer: Hypothetical Document Embeddings uses a generated hypothetical answer or document to improve retrieval matching.

Q. What is GraphRAG?

Answer: GraphRAG combines graph-based relationships with retrieval to help answer questions requiring entity and relationship reasoning.

Q. What is RAG grounding?

Answer: Grounding means generating answers using retrieved evidence rather than relying only on model memory.

Q. What causes poor RAG answers?

Answer: Common causes include bad parsing, poor chunking, weak retrieval, missing metadata filters, and insufficient context.

Q. How do you prevent unauthorized RAG access?

Answer: Apply authorization before retrieval and enforce tenant or document-level access filters.

Q. What is RAG evaluation?

Answer: It measures retrieval quality, answer correctness, faithfulness, relevance, and other task-specific metrics.

6. AI Agents & Agentic AI

Q. What is an AI agent?

Answer: An AI agent is a system that uses a model, instructions, tools, state, and feedback to pursue a goal.

Q. How is an agent different from a chatbot?

Answer: A chatbot mainly responds conversationally, while an agent can plan, use tools, and take actions toward a goal.

Q. What is tool use?

Answer: Tool use lets a model invoke external capabilities such as search, databases, APIs, or code execution.

Q. What is an agent loop?

Answer: A typical loop is observe context, decide or plan, call a tool, inspect the result, and continue until completion.

Q. What is agent memory?

Answer: Memory stores useful information across steps or interactions so an agent can maintain context.

Q. What is short-term memory?

Answer: Short-term memory holds information needed within the current task or conversation.

Q. What is long-term memory?

Answer: Long-term memory persists selected information across sessions or tasks.

Q. What is human-in-the-loop?

Answer: A human reviews, approves, or intervenes in selected agent decisions or actions.

Q. Why is human approval important?

Answer: It reduces risk for sensitive actions such as financial transactions, production changes, or destructive operations.

Q. What is a single-agent system?

Answer: One agent coordinates reasoning and tool use to complete a task.

Q. What is a multi-agent system?

Answer: Multiple specialized agents collaborate or delegate parts of a larger task.

Q. What is agent orchestration?

Answer: Orchestration controls how agents, tools, state, routing, retries, and approvals work together.

Q. What is planning in an agent?

Answer: Planning decomposes a goal into steps or actions that can be executed and revised.

Q. What is plan-and-execute?

Answer: A planner creates a sequence of tasks and an executor performs them, sometimes with replanning.

Q. What is agent state?

Answer: State is the structured information an agent carries across execution steps.

Q. What is a guardrail?

Answer: A guardrail is a rule or control that restricts unsafe inputs, outputs, actions, or tool usage.

Q. What is an agent handoff?

Answer: A handoff transfers a task from one specialized agent to another.

Q. What is agent retry logic?

Answer: Retry logic reattempts transient failures using controlled limits and often exponential backoff.

Q. What is an agent trace?

Answer: A trace records model calls, tool calls, decisions, latency, errors, and other execution events for debugging.

Q. What is agent evaluation?

Answer: It measures whether an agent completes tasks correctly, safely, efficiently, and consistently.

7. MCP, APIs & AI Integration

Q. What is MCP?

Answer: Model Context Protocol is an open protocol that standardizes how AI applications connect models with tools and context.

Q. Why is MCP useful for agents?

Answer: It provides a standardized way to expose tools and resources so agents can interact with external systems.

Q. What is an MCP server?

Answer: An MCP server exposes tools, resources, or prompts that an AI application can discover and use.

Q. What is an MCP client?

Answer: An MCP client connects an AI application or agent to an MCP server.

Q. What is an MCP host?

Answer: The host is the AI application that manages the user interaction and one or more MCP client connections.

Q. What is an MCP tool?

Answer: An MCP tool is a callable operation exposed by a server for an AI application to invoke.

Q. What is an MCP resource?

Answer: An MCP resource is contextual data that a client can retrieve from an MCP server.

Q. What is MCP stdio transport?

Answer: Stdio connects a client to a local MCP server process through standard input and output.

Q. What is Streamable HTTP in MCP?

Answer: It is an HTTP-based transport designed for communicating with MCP servers over a network.

Q. Why should MCP servers be trusted carefully?

Answer: Tools can expose data or perform actions using supplied credentials, so least privilege and approval controls are important.

Q. What is an API?

Answer: An API is a defined interface through which software components communicate.

Q. What is REST?

Answer: REST is an architectural style commonly used for HTTP-based resource-oriented APIs.

Q. What is JSON?

Answer: JSON is a lightweight text format for representing structured data using objects, arrays, and primitive values.

Q. What is YAML?

Answer: YAML is a human-friendly serialization format commonly used for configuration and structured data.

Q. What is authentication?

Answer: Authentication verifies who a user, service, or application is.

Q. What is authorization?

Answer: Authorization determines what an authenticated identity is allowed to access or do.

Q. What is OAuth?

Answer: OAuth is an authorization framework that allows delegated access without sharing a user's password with the client.

Q. What is least privilege?

Answer: Least privilege gives an identity only the minimum access required to perform its task.

8. LLM Evaluation & Observability

Q. Why is LLM evaluation difficult?

Answer: LLM outputs can be variable, open-ended, and context-dependent, so quality cannot be captured by one metric.

Q. What is a golden dataset?

Answer: A curated set of representative inputs and expected or reference outcomes used for evaluation.

Q. What is offline evaluation?

Answer: It evaluates models or pipelines against a fixed dataset before production deployment.

Q. What is online evaluation?

Answer: It measures quality and behavior using real production traffic or controlled live experiments.

Q. What is accuracy?

Answer: Accuracy is the fraction of predictions that are correct among all evaluated predictions.

Q. What is precision?

Answer: Precision measures how many predicted positives are actually positive.

Q. What is recall?

Answer: Recall measures how many actual positives the system successfully identifies.

Q. What is F1 score?

Answer: F1 is the harmonic mean of precision and recall.

Q. What is faithfulness in RAG?

Answer: Faithfulness measures whether the generated answer is supported by the supplied evidence.

Q. What is answer relevance?

Answer: Answer relevance measures whether the response directly addresses the user's question.

Q. What is retrieval recall?

Answer: Retrieval recall measures how often relevant evidence is included among retrieved candidates.

Q. What is latency?

Answer: Latency is the time taken for a request or a stage of processing to complete.

Q. What is throughput?

Answer: Throughput is the amount of work or requests processed per unit of time.

Q. What is token usage?

Answer: Token usage is the amount of input and output token processing consumed by a model call.

Q. How do you reduce LLM cost?

Answer: Use smaller models where appropriate, control context size, cache results, optimize prompts, and monitor token usage.

Q. What is LLM observability?

Answer: It is monitoring model quality, traces, latency, cost, errors, safety signals, and user outcomes.

Q. What is prompt versioning?

Answer: It is tracking prompt changes so behavior can be reproduced, compared, and rolled back.

9. AI Safety, Security & Responsible AI

Q. What is responsible AI?

Answer: Responsible AI aims to build systems that are safe, fair, transparent, accountable, privacy-aware, and reliable.

Q. What is AI bias?

Answer: AI bias is systematic skew in data, model behavior, or decisions that can produce unfair outcomes.

Q. What is model safety?

Answer: Model safety focuses on preventing harmful, unsafe, or unauthorized behavior.

Q. What is data privacy?

Answer: Data privacy concerns appropriate collection, use, storage, sharing, and protection of personal information.

Q. What is PII?

Answer: Personally Identifiable Information is information that can identify or reasonably be linked to an individual.

Q. What is data minimization?

Answer: Data minimization means collecting and processing only the information necessary for the intended purpose.

Q. What is prompt injection defense?

Answer: Use instruction/data separation, input handling, tool permissions, validation, sandboxing, and human approval for risky actions.

Q. What is jailbreak?

Answer: A jailbreak is an attempt to bypass a model's safety or behavioral restrictions.

Q. What is output validation?

Answer: Output validation checks model-generated content against schemas, policies, or business rules before use.

Q. What is a sandbox?

Answer: A sandbox isolates code or actions to limit their access and potential impact.

Q. What is content moderation?

Answer: Content moderation detects, blocks, transforms, or routes content according to safety policies.

Q. What is explainability?

Answer: Explainability is the ability to provide understandable reasons or evidence for a model's behavior.

Q. What is auditability?

Answer: Auditability means maintaining records that allow decisions and system actions to be reviewed.

Q. What is AI governance?

Answer: AI governance defines policies, controls, ownership, risk management, evaluation, and lifecycle processes for AI systems.

Q. What is model governance?

Answer: Model governance manages model approval, versions, risk, monitoring, documentation, and retirement.

10. Deployment, MLOps & LLMOps

Q. What is MLOps?

Answer: MLOps applies software engineering, automation, monitoring, and governance practices to machine-learning lifecycles.

Q. What is LLMOps?

Answer: LLMOps adapts operational practices for LLM applications, including prompts, retrieval, evaluation, inference, and safety.

Q. What is CI/CD?

Answer: CI/CD automates integration, testing, packaging, and deployment of software changes.

Q. What is model serving?

Answer: Model serving exposes a trained model so applications can send inference requests.

Q. What is an inference endpoint?

Answer: It is an API endpoint through which clients submit inputs and receive model predictions or generations.

Q. What is autoscaling?

Answer: Autoscaling adjusts compute capacity based on workload demand.

Q. What is caching in AI applications?

Answer: Caching stores reusable results or artifacts to reduce latency, compute, and cost.

Q. What is batching?

Answer: Batching processes multiple requests together to improve hardware utilization and throughput.

Q. What is streaming inference?

Answer: Streaming sends generated output incrementally instead of waiting for the complete response.

Q. What is vLLM?

Answer: vLLM is an inference and serving engine designed to efficiently serve large language models.

Q. What is KV cache?

Answer: KV cache stores attention key/value states from prior tokens to accelerate autoregressive generation.

Q. What is speculative decoding?

Answer: A smaller or faster model proposes tokens that a larger model verifies, potentially reducing generation latency.

Q. What is model distillation?

Answer: Distillation trains a smaller model to reproduce useful behavior from a larger teacher model.

Q. What is model routing?

Answer: Model routing selects an appropriate model based on factors such as task, quality, latency, and cost.

Q. What is canary deployment?

Answer: Canary deployment releases a change to a small portion of traffic before broader rollout.

Q. What is rollback?

Answer: Rollback returns a system to a previously known-good version after a problematic release.

Q. What is drift?

Answer: Drift is a meaningful change in input data, behavior, or performance distribution over time.

11. Cloud, Data & Architecture

Q. What is an AI inference architecture?

Answer: It is the system design that receives requests, prepares context, invokes models/tools, validates outputs, and returns results.

Q. What is a data pipeline?

Answer: A data pipeline moves and transforms data through ingestion, processing, storage, and consumption stages.

Q. What is ETL?

Answer: ETL means Extract, Transform, Load.

Q. What is a data lake?

Answer: A data lake stores large volumes of raw or semi-structured data for flexible processing.

Q. What is a data warehouse?

Answer: A data warehouse stores structured, curated data optimized for analytics.

Q. What is a feature store?

Answer: A feature store manages reusable ML features for training and inference.

Q. What is object storage?

Answer: Object storage stores files or blobs as objects, typically with metadata and unique keys.

Q. Why use a message queue in AI systems?

Answer: Queues decouple components and support asynchronous, resilient processing.

Q. What is asynchronous processing?

Answer: A request can trigger work that completes later without blocking the caller for the entire operation.

Q. What is a microservice?

Answer: A microservice is a small independently deployable service focused on a specific capability.

Q. What is an API gateway?

Answer: An API gateway provides a common entry point for routing, authentication, throttling, and policy enforcement.

Q. What is rate limiting?

Answer: Rate limiting controls how frequently a client can make requests.

Q. What is multi-tenancy?

Answer: Multi-tenancy lets one system serve multiple customers while isolating their data and access.

Q. Why is tenant isolation important in RAG?

Answer: It prevents one customer's documents or retrieved context from being exposed to another customer.

Q. What is horizontal scaling?

Answer: Horizontal scaling adds more service instances to handle increased workload.

12. Practical Scenario Questions

Q. How would you build an enterprise document chatbot?

Answer: Use document ingestion, chunking, embeddings, secure vector retrieval, reranking, an LLM, citations, access controls, and evaluation.

Q. How would you reduce hallucinations in a chatbot?

Answer: Use grounded retrieval, explicit instructions, constrained generation, validation, citations, and abstention when evidence is insufficient.

Q. How would you handle a large PDF in RAG?

Answer: Parse it reliably, preserve metadata and structure, chunk intelligently, embed chunks, and retrieve relevant sections.

Q. How would you support multiple document types?

Answer: Use format-specific parsers followed by a common normalized document representation.

Q. How would you handle stale knowledge?

Answer: Refresh the knowledge source and index on a controlled schedule or event-driven pipeline.

Q. How would you protect confidential documents?

Answer: Enforce identity-aware authorization and metadata filters before retrieval and generation.

Q. How would you choose an LLM?

Answer: Compare task quality, context needs, latency, cost, tool support, safety, deployment constraints, and evaluation results.

Q. How would you choose an embedding model?

Answer: Benchmark retrieval quality on representative domain queries while considering language coverage, latency, and cost.

Q. How would you debug a poor RAG answer?

Answer: Inspect ingestion, chunking, query formulation, retrieved documents, reranking, context construction, prompt, and generation.

Q. How would you debug an agent that loops forever?

Answer: Add step limits, tool-call limits, state checks, timeouts, loop detection, and clear termination criteria.

Q. How would you make an agent safe for production?

Answer: Use least-privilege tools, validation, guardrails, sandboxing, tracing, rate limits, approvals, and extensive evaluation.

Q. How would you reduce agent latency?

Answer: Reduce unnecessary model calls, parallelize independent tools, cache stable data, use smaller models, and optimize retrieval.

Q. How would you reduce agent cost?

Answer: Limit context and steps, route simple tasks to cheaper models, cache results, and monitor token and tool usage.

Q. How would you handle tool failure?

Answer: Return structured errors, retry transient failures with backoff, fall back when possible, and stop safely when recovery fails.

Q. How would you evaluate an AI agent?

Answer: Use task success, correctness, safety, tool accuracy, latency, cost, robustness, and human review where appropriate.

Q. How would you design human approval for an agent?

Answer: Require approval before predefined high-impact actions while allowing low-risk actions to proceed automatically.

Q. How would you prevent an LLM from executing arbitrary code?

Answer: Use restricted tools, allowlists, sandboxing, resource limits, and explicit approval for sensitive operations.

Q. How would you handle prompt injection in retrieved documents?

Answer: Treat retrieved content as untrusted data and prevent it from overriding system instructions or gaining uncontrolled tool access.

Q. How would you design a production RAG system?

Answer: Separate ingestion and serving, secure retrieval, cache embeddings/results, add observability, evaluation, scaling, and disaster recovery.

Q. What would you monitor after deploying an LLM application?

Answer: Monitor quality, hallucination/faithfulness signals, latency, errors, token cost, retrieval quality, safety events, and user outcomes.